Haocheng Xi

Berkeley AI Research, University of California, Berkeley | xihc@berkeley.edu
haochengxi.github.io

EDUCATION

University of California, Berkeley — Berkeley, CA
Ph.D. in Computer Science, Berkeley AI Research (BAIR) — 09/2024 – Present
Advisor: Prof. Kurt Keutzer

Tsinghua University — Beijing, China
B.Eng. in Computer Science & Technology — 09/2020 – 06/2024
Institute for Interdisciplinary Information Sciences (IIIS)
Yao Class, led by Prof. Andrew C.C. Yao

University of Washington — Seattle, WA
Visiting Student, Paul G. Allen School of Computer Science & Engineering — 02/2023 – 08/2023
Advisor: Prof. Sheng Wang

Beijing No.8 High School — Beijing, China
Experimental class for gifted and talented young, Excellent Graduate — 09/2015 – 07/2020

RESEARCH INTERESTS

My research interests lie in efficient machine learning, such as quantization and sparsity. I aim to push the boundaries of how we can effectively compress and accelerate deep learning models while maintaining their accuracy and robustness.

SELECTED PUBLICATIONS

Efficient Video Generation

Video DeltaNet: A Video-Native Hybrid Attention for Livestream Video Generation

Haocheng Xi, Yiming Xie, Hexu Zhao, Yiwen Zhang, Michael Liu, Thomas Creavin, Kurt Keutzer, Xiuyu Li, Zhaoyang Lv, Chenfeng Xu, Haiwen Feng
arXiv preprint, 2026. [arxiv] [code] [website]

Quant VideoGen: Auto-Regressive Long Video Generation via 2-Bit KV-Cache Quantization

Haocheng Xi, Shuo Yang*, Yilong Zhao, Muyang Li, Han Cai, Xingyang Li, Yujun Lin, Zhuoyang Zhang, Jintao Zhang, Xiuyu Li, Zhiying Xu, Jun Wu, Chenfeng Xu, Ion Stoica, Song Han, Kurt Keutzer*
International Conference on Machine Learning (ICML), 2026. [arxiv] [code]

Sparse VideoGen: Accelerating Video Diffusion Transformers with Spatial-Temporal Sparsity

Haocheng Xi, Shuo Yang*, Yilong Zhao, Chenfeng Xu, Muyang Li, Xiuyu Li, Yujun Lin, Han Cai, Jintao Zhang, Dacheng Li, Jianfei Chen, Ion Stoica, Kurt Keutzer, Song Han*
International Conference on Machine Learning (ICML), 2025. [arxiv] [code] [website] [poster]

Flash-KMeans: Fast and Memory-Efficient Exact K-Means

Shuo Yang, Haocheng Xi*, Yilong Zhao, Muyang Li, Xiaoze Fan, Jintao Zhang, Han Cai, Yujun Lin, Xiuyu Li, Kurt Keutzer, Song Han, Chenfeng Xu, Ion Stoica*
Conference on Neural Information Processing Systems (NeurIPS), 2026 (accepted). [arxiv] [code]

Sparse VideoGen2: Accelerate Video Generation with Sparse Attention via Semantic-Aware Permutation

Shuo Yang, Haocheng Xi*, Yilong Zhao, Muyang Li, Jintao Zhang, Han Cai, Yujun Lin, Xiuyu Li, Chenfeng Xu, Kelly Peng, Jianfei Chen, Song Han, Kurt Keutzer, Ion Stoica*
Conference on Neural Information Processing Systems (NeurIPS), 2025. [arxiv] [code] [website]
Selected as **Spotlight Paper** in NeurIPS 2025.

MosaiChunk: Compositing Spatio-Temporal Memory for Autoregressive Video Generation

Yiwen Zhang, Haocheng Xi, Michael Tian-Yue Liu, Alexei A. Efros, Hadar Averbuch-Elor, Qianqian Wang, Haiwen Feng
arXiv preprint, 2026. [arxiv] [code] [website]

StreamDiffusionV2: A Streaming System for Dynamic and Interactive Video Generation

Tianrui Feng, Zhi Li, Shuo Yang, Haocheng Xi, Muyang Li, Xiuyu Li, Lvmin Zhang, Keting Yang, Kelly Peng, Song Han, Maneesh Agrawala, Kurt Keutzer, Akio Kodaira, Chenfeng Xu
Conference on Machine Learning and Systems (MLSys), 2026. [arxiv] [code] [website]
Recipient of **Best Research Paper Award** at MLSys 2026.

Efficient Language Models

LoSA: Locality Aware Sparse Attention in Diffusion Language Models

Haocheng Xi, Harman Singh, Yuezhou Hu, Coleman Hooper, Rishabh Tiwari, Aditya Tomar, Minjae Lee, Wonjun Kang, Michael W. Mahoney, Chenfeng Xu, Kurt Keutzer, Amir Gholami
International Conference on Machine Learning (ICML), 2026. [arxiv] [proceedings]

Jet-RL: Enabling On-Policy FP8 Reinforcement Learning with Unified Training and Rollout Precision Flow

Haocheng Xi, Charlie Ruan, Peiyuan Liao, Yujun Lin, Han Cai, Yilong Zhao, Shuo Yang, Kurt Keutzer, Song Han, Ligeng Zhu
arXiv preprint, 2026. [arxiv]

COAT: Compressing Optimizer states and Activations for Memory-Efficient FP8 Training
Haocheng Xi, Han Cai, Ligeng Zhu, Yao Lu, Kurt Keutzer, Jianfei Chen, Song Han
International Conference on Learning Representations (ICLR), 2025. [arxiv] [code] [website]

Jetfire: Efficient and Accurate Transformer Pretraining with INT8 Data Flow and Per-Block Quantization

Haocheng Xi, Yuxiang Chen, Kang Zhao, Kai Jun Teh, Jianfei Chen, Jun Zhu
International Conference on Machine Learning (ICML), 2024. [arxiv] [code] [poster]
Selected as **Spotlight Paper** in ICML 2024.

Training Transformers with 4-bit Integers

Haocheng Xi, Changhao Li, Jianfei Chen, Jun Zhu
Conference on Neural Information Processing Systems (NeurIPS), 2023. [arxiv] [code]

LeapQuant: Efficient Linear Attention with Accurate Recurrent State Quantization

Yi Pan, Haocheng Xi*, Kan Zhu, Xingyang Li, Yibo Wu, Mayank Mishra, Hongtao Zhang, William X. Zheng, Baris Kasikci, Song Han, Kurt Keutzer, Rishabh Iyer, Ion Stoica*
arXiv preprint, 2026. [arxiv]

QuantSpec: Self-Speculative Decoding with Hierarchical Quantized KV Cache

Rishabh Tiwari, Haocheng Xi*, Aditya Tomar*, Coleman Hooper, Sehoon Kim, Maxwell Horton, Mahyar Najibi, Michael W. Mahoney, Kurt Keutzer, Amir Gholami*
International Conference on Machine Learning (ICML), 2025. [arxiv] [poster]

Oscillation-Reduced MXFP4 Training for Vision Transformers

Yuxiang Chen, Haocheng Xi, Jun Zhu, Jianfei Chen
International Conference on Machine Learning (ICML), 2025. [arxiv]

INTERNSHIP EXPERIENCE

Stealth Company, Intern 05/2026 – 08/2026

NVIDIA Research, Research Intern 05/2025 – 04/2026

NVIDIA Research, Research Intern 03/2024 – 08/2024

HONORS

Citadel Fellowship, In 2026
Fellowship of Tsinghua Xuetang Talents Program Among top 300 / 3000 Tsinghua students each year
Athletic Excellence Scholarship In 2022
First Prize of National Senior High School Mathematics Competition In 2019

SKILLS

Language: TOFEL: Total 110 (Reading 29, Listening 29, Speaking 24, Writing 28)
GRE: Quantitive 170, Verbal 158, Writing 4.0
Programming and Software: Python, CUDA, C++, Bash, Git, LaTeX
Deep Learning Package: PyTorch, Transformers, Triton, PEFT, TransformerEngine, VeRL
Conference review: ICML (2025), NeurIPS (2024, 2025), ICLR (2025, 2026)